

The Relationship Between Loneliness and Insomnia is Mediated by TV Binge-Watching

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Abstract

Research suggests that loneliness is associated with insomnia. Because binge-watching has been found to correlate with both loneliness and insomnia, we examined whether it acts as a moderator or mediator of the loneliness-insomnia relationship. Using a sample of 447 undergraduates, binge-watching was shown to mediate, but not moderate, the relationship.

Introduction

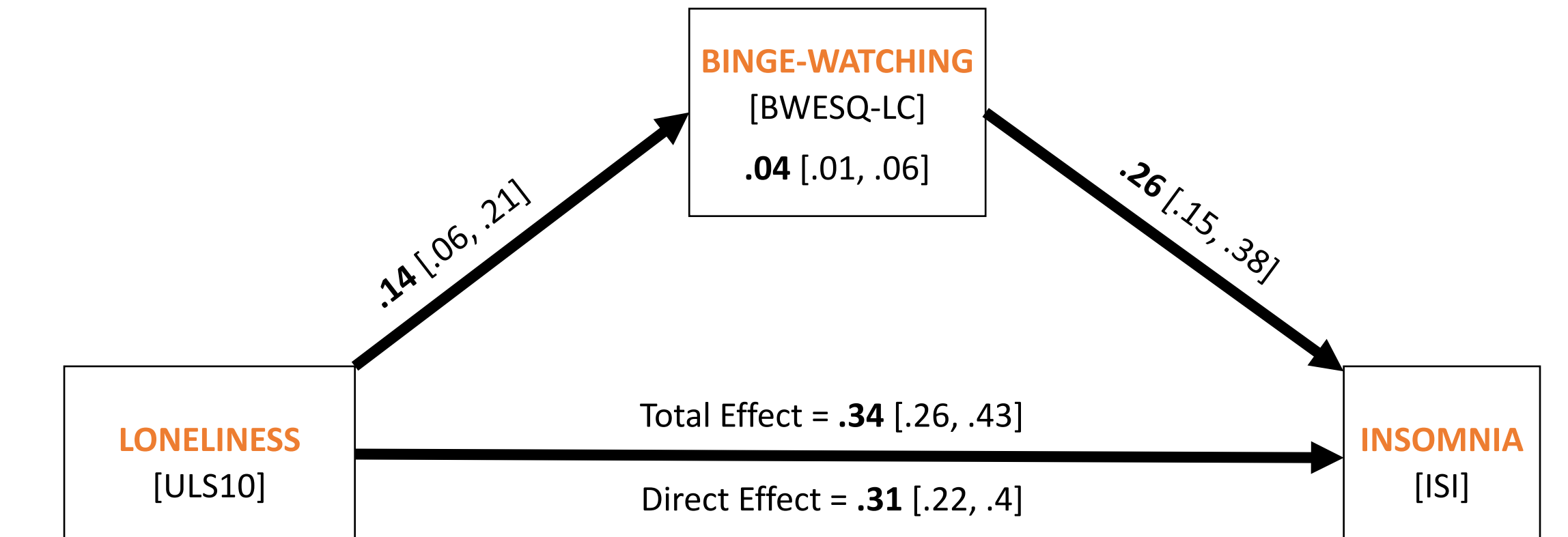
Inadequate sleep has been shown to be associated with physical health issues, including hypertension risk and elevated blood pressure (Makarem, 2019). It is important therefore to identify factors that might contribute to inadequate sleep. A number of studies have suggested that loneliness may be a contributing factor, having found to be associated with sleep problems, including insomnia (Griffin et al., 2020). It is important to identify factors that might influence (moderate) the strength of this loneliness-insomnia relationship and factors that might explain the mechanism through which loneliness affects insomnia (mediators). Because prior research has shown an association between loneliness and binge-watching (Sun et al., 2021; Alfonsi et al., 2023) and between binge-watching and poor sleep (Jindal, 2020; Alfonsi et al., 2023), the purpose of this study was to examine whether TV binge-watching acted as a moderator or mediator of the loneliness-insomnia relationship.

Methods

Four hundred and forty-seven undergraduates (73.2% female; 95.1% Hispanic; mean age 20.4) anonymously completed a Qualtrics survey. The survey's measures included the Insomnia Severity Index, a "binge-watching factor from the Binge-Watching Engagement and Symptoms Questionnaire, and a 10-item version of the UCLA Loneliness Scale (Ver. 3) (Russell, 1996). In order to examine whether binge-watching served as a significant moderator or mediator of the loneliness-insomnia relationship, separate moderation and mediation analyses were conducted using Hayes' PROCESS Macro V4.1 with 5,000 bootstrapped samples, controlling for both age and sex.

Results

Loneliness was positively correlated with insomnia ($r(445) = .35, p < .001$). Binge-watching did not significantly moderate this association (interaction: $b = -.002, t(439) = -.23, p > .05, CI[-.021, .016]$). However, binge-watching significantly mediated the loneliness-insomnia relationship (unstandardized indirect effect = 0.04, $SE = 0.01, 95\% CI [0.01, 0.06]$). 17% of the variance in insomnia severity scores was predicted by this model.



Conclusion

We replicated prior research showing an association between loneliness and insomnia. Our replication in a predominantly Hispanic sample is important given the limited inclusion of Hispanics in prior studies, with only 5.3% including Hispanic/Latinx as an identifier (Hom et al., 2020). The nonsignificant moderation suggests that the strength of the relationship between loneliness and insomnia is not influenced by the amount of binge-watching a person engages in. However, our findings indicate that binge-watching might partially explain the loneliness-insomnia relationship, serving as one potential mechanism connecting the two experiences. Our findings tentatively suggest that interventions to reduce binge-watching be considered as a means of reducing the impact of loneliness on insomnia. However, a limitation of our study is that it was cross-sectional and so cannot be used as evidence of causal relationships. It is also possible that relationships between these variables could be bidirectional, creating a vicious cycle of behavior. Future studies should consider longitudinal designs to address such questions.

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